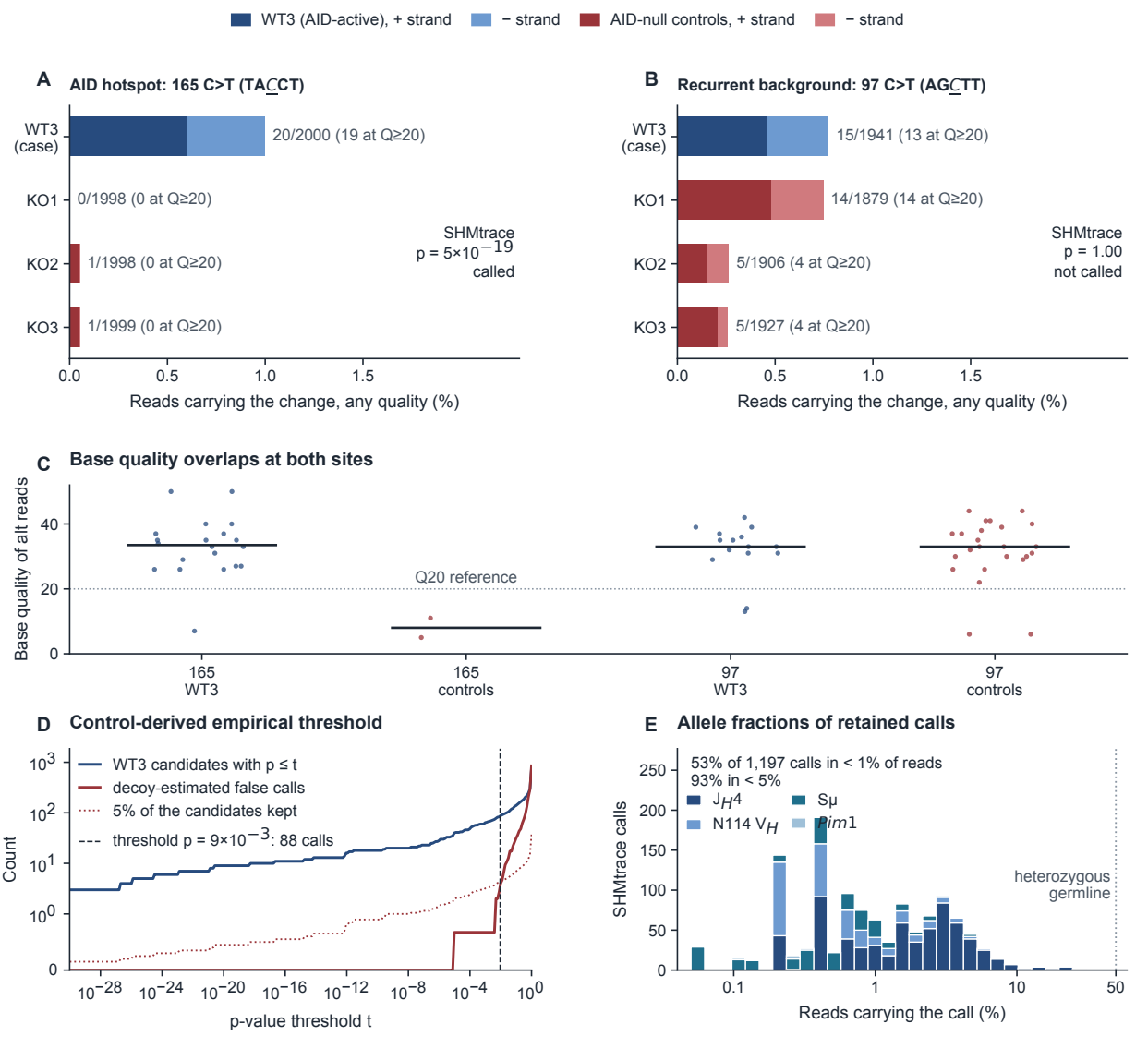

WT3 (AID-active), + strand
− strand
AID-null controls, + strand
− strand
A
AID hotspot: 165 C>T (TACCT)
WT3 (case)
KO1
KO2
KO3
20/2000 (19 at Q≥20)
0/1998 (0 at Q≥20)
1/1998 (0 at Q≥20)
1/1999 (0 at Q≥20)
SHMtrace
p = 5×10−19
called
0.0
0.5
1.0
1.5
Reads carrying the change, any quality (%)
B
Recurrent background: 97 C>T (AGCTT)
WT3 (case)
KO1
KO2
KO3
15/1941 (13 at Q≥20)
14/1879 (14 at Q≥20)
5/1906 (4 at Q≥20)
5/1927 (4 at Q≥20)
SHMtrace
p = 1.00
not called
0.0
0.5
1.0
1.5
Reads carrying the change, any quality (%)
C
Base quality overlaps at both sites
Base quality of alt reads
0
20
40
Q20 reference
165 WT3
165 controls
97 WT3
97 controls
D
Control-derived empirical threshold
Count
10^0
10^1
10^2
10^3
WT3 candidates with p ≤ t
decoy-estimated false calls
5% of the candidates kept
threshold p = 9×10−3: 88 calls
10−28
10−24
10−20
10−16
10−12
10−8
10−4
100
p-value threshold t
E
Allele fractions of retained calls
53% of 1,197 calls in < 1% of reads
93% in < 5%
JH4
Sμ
N114 VH
Pim1
SHMtrace calls
0
50
100
150
200
250
heterozygous germline
0.1
1
10
50
Reads carrying the call (%)